Forensic Science, Medicine and Pathology (2023) 19:557–562
https://doi.org/10.1007/s12024-022-00570-6

CASE REPORT

Inferior vena cava injury during percutaneous vertebroplasty: a rare cause of fatal pulmonary thromboembolism

Qian Wang[1] · Min Yin[1] · Qing Shi[1] · Yijie Duan[1] · Shuquan Zhao[1] · Yiwu Zhou[1]

Accepted: 21 December 2022 / Published online: 28 December 2022

Abstract

Fatal pulmonary thromboembolism (PTE) following percutaneous vertebroplasty is rare in medical practice. Here, we report the case of a 70-year-old woman who suffered from lumbago with lower extremity pain and lameness and for whom lumbar osteoporotic vertebral compression fractures (L4, L5) were seen on MRI examination. Percutaneous vertebroplasty and posterior vertebral lamina fenestration discectomy were performed. One day later, her condition deteriorated after defecation, and she died suddenly. Pulmonary thromboembolus and deep venous thrombosis in the inferior vena cava were the major findings at forensic autopsy. Due to the rather uncommon components of the thromboembolus (chondrocytes, calcium deposits, and collagen fibers), the pulmonary thromboembolism was attributed to deep venous thrombosis in the inferior vena cava, which was injured during percutaneous vertebroplasty. The present study highlights the conclusion that pulmonary thromboembolism is a rare complication of percutaneous vertebroplasty that should attract the attention of clinical physicians and forensic pathologists.

Keywords Forensic Pathology · Pulmonary thromboembolism · Percutaneous vertebroplasty · Inferior vena cava injure · Medical malpractice

Introduction

Vertebral compression fractures (VCFs) are one of the most common types of fractures in patients with osteoporosis and are directly correlated with age and sex [1]. In 1984, Mansour et al. introduced percutaneous vertebroplasty (PVP) to treat VCFs. Since then, the procedure has become popular because it relieves pain and is easy to perform [2]. The incidence of its complications is approximately 1–5%, and its mortality rate is less than 1%. Bone cement leakage is the leading cause of adverse events. Fatal pulmonary failure occurring immediately postoperatively is uncommon, and it has been reported that most cases are caused by polymethylmethacrylate (PMMA) or fat embolism [3–6]. Nevertheless, fatal pulmonary thromboembolism (PTE) secondary to PVP is rather rare in clinical and forensic practice. PTE occurs when a part of the thrombus breaks loose from somewhere in the venous circulation, then migrates and subsequently becomes lodged in the pulmonary arteries. Here, we present a 70-year-old female suffering from osteoporosis with pathologic lumbar compression fractures (L4, L5) who died 5 days after percutaneous vertebroplasty. Pulmonary thromboembolus and deep venous thrombosis in the inferior vena cava were the major findings at forensic autopsy. Due to the rather uncommon components of the embolisms (chondrocytes, calcium deposits, and collagen fibers), the pulmonary thromboembolism was attributed to deep venous thrombosis in the inferior vena cava, which was injured during percutaneous vertebroplasty.

Qian Wang and Min Yin contributed equally to this work.

✉ Shuquan Zhao
fyzsq123@126.com

✉ Yiwu Zhou
zhouyiwu@outlook.com

1 Department of Forensic Medicine, Tongji Medical College, Huazhong University of Science and Technology, No. 13 Hangkong Road, Wuhan 430030, People's Republic of China

Clinical history

A 70-year-old woman presented at the hospital with a complaint of lumbago with lower extremity pain and lameness lasting for 1 month. Cerebral infarction had been diagnosed in 2020 without any treatment. Physical examination by a spine specialist revealed mild kyphosis of the thoracolumbar spine and tenderness on both sides of the lumbar spinous process. The pain was aggravated when the back was extended, and decreased skin sensation was observed on the medial malleolus, the outer side of the calf, and the dorsum of the foot of the right lower extremity. The strength of the muscle scored grade III on the dorsal extensor of the right lower extremity ankle and extensor hallucis dorsi, and absent knee tendon reflexes were also seen in both lower limbs. MRI of the lumbar spine showed osteoporosis with pathologic lumbar compression fractures (L4, L5) and lumbar disc herniation with spinal stenosis (L4-5, L5-S1) (Fig. 1). Her past medical history was unremarkable. The coagulation function test results were as follows: PT-sec, 11.2 s; PT-INR, 0.97; PT%, 98%; APTT, 23.0 s; FIB, 2.57 g/L; and TT2, 18.90 s.

Percutaneous vertebroplasty was performed at L4-L5 on Day 4 in the hospital. After anesthesia, the fracture cones (L4 and L5) were determined under fluoroscopy. A needle was used to enter the vertebral body through the pedicle on the right side of L4 and the left side of L5, and 4 ml of PMMA was injected directly into the fractured vertebral body. Fluoroscopy showed that the bone cement adequately filled the vertebral body. The operation lasted 4 h. During the operation, the patient's vital signs were stable. After the operation, the patient's back pain was relieved, and she was able to get out of bed for activities. However, the pain in her legs was aggravated after standing.

Four days later, a posterior vertebral lamina fenestration discectomy was performed at L5-S1. After anesthesia, L5 right lamina fenestration was performed routinely. The posterior longitudinal ligament located in front of the L5-S1 intervertebral disc was excised. Then, a portion of the intervertebral disc tissues was removed with nucleus pulposus forceps. The operation lasted 2.5 h. After the operation, the patient's leg pain was significantly relieved. However, 1 day after the operation, the patient felt dizzy when getting out of bed to defecate. She suddenly lost consciousness and died.

Forensic autopsy

Due to the sudden postoperative death of the decedent, medical malpractice was suspected by her family, and a forensic autopsy was performed 20 h later.

Except for two surgical wounds that were 0.3 cm long on the right side of L4 and 4.0 cm long in the middle of L5 on the back, no injuries were found during the external examination.

The right ventricular outflow tract and the pulmonary artery were incised, and a saddle pulmonary embolism (PE) in the pulmonary artery was seen. The thromboembolus was red, white, and firm. The pulmonary hilar arteries and their branches were obstructed by the thromboembolus (Fig. 2a).

The left and right common iliac veins and inferior vena cava were opened, and a 1.7 cm × 1.2 cm red, white, and firm thrombus was attached to the intima of the inferior vena cava at the level of L5 (Fig. 2b). Separating the thrombus, a 1.5 cm×1.0 cm defect in the vessel wall on the right side of the L5 vertebral body was found where the thrombus was located. A hole-like defect was located on the right side of the vertebral body, and the surrounding soft tissue was bleeding. After opening the dorsal surgical incision, a probe was passed through the L5 surgical incision and the L5 vertebral body to the inferior vena cava defect (Fig. 3). The intervertebral disc between L4 and L5 was sawed open. The intervertebral disc was observed to be relatively intact, and a surgical wound was seen on the transverse section. The wound passed through part of the intervertebral disc from the pedicle of L5 and finally penetrated the L5 vertebral body. Around the wound, there was an irregular, milky yellow bone cement, which could be easily separated from the surrounding bone (Fig. 4). Opening the spinal cavity, small

Fig. 1 MRI of the lumbar spine showed lumbar osteoporotic vertebral compression fractures (L4, L5) and lumbar disc herniation with spinal stenosis (L4-5, L5-S1)

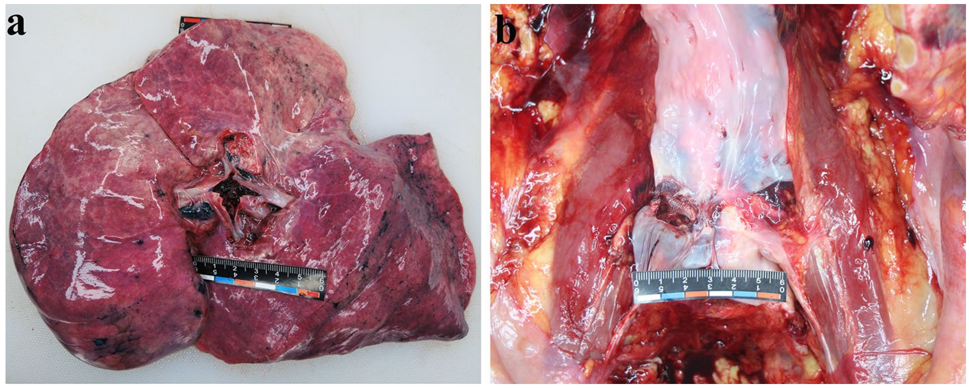

Fig. 2 Macroscopic pictures of the lesions. **a** The pulmonary hilar arteries and their branches were found to be obstructed by the thromboembolism. **b** A thrombus was attached to the intima of the inferior vena cava

hematomas were scattered throughout the lumbar epidural region. Both lower extremities were incised with negative findings in the deep veins.

The thromboembolus in the pulmonary artery and the thrombus in the inferior vena cava displayed mixed thrombi under a microscope (Fig. 5a). Chondrocytes, calcium deposits (bone), and collagen fibers (fascia) were the uncommon components of the thromboembolus (Fig. 5b, c). Masson staining was performed, and blue collagen fibers were observed in the hilar thromboembolus (Fig. 5d). The rather uncommon components of the pulmonary thromboembolism and deep venous thrombosis confirmed their common origin. It was determined that these components came from the bone injury that occurred during the operation.

Multiple hemorrhages and calcium deposits (bone) were seen in the right defect of the L5 vertebral body. Hemorrhage with neutrophil infiltration was observed at the rupture of the inferior vena cava and L5 vertebral body.

In this case, pulmonary thromboembolism, originating from deep venous thrombosis in the inferior vena cava, was caused by PVP performed to treat osteoporosis with pathological lumbar vertebral compression fractures, which was identified as the cause of death.

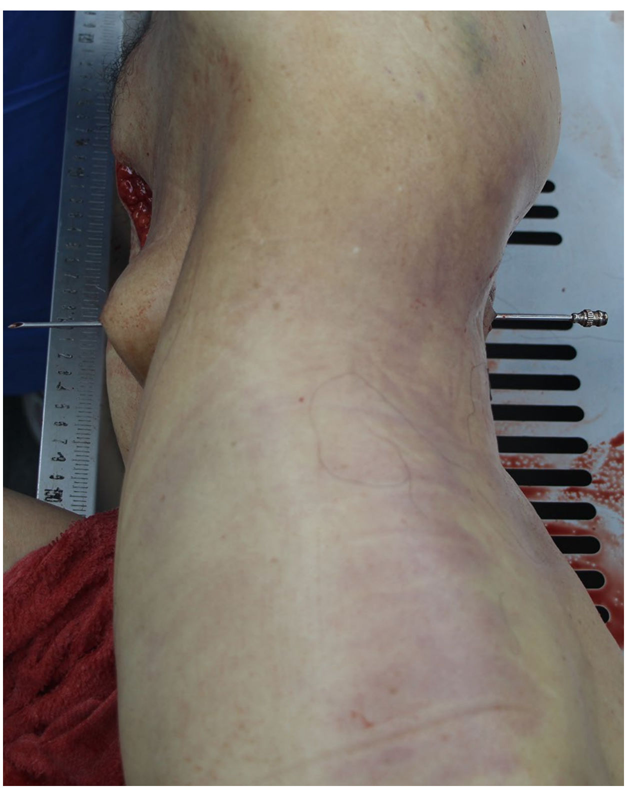

Fig. 3 The picture shows a probe that was passed through the L5 surgical incision and then through the L5 vertebral body perforation to the inferior vena cava breach

Discussion

According to the literature, 65% of PVP cases have bone cement leakage during or after the procedure. The cement enters the venous system or the spinal canal, and most patients are asymptomatic [7]. Bone cement leakage usually

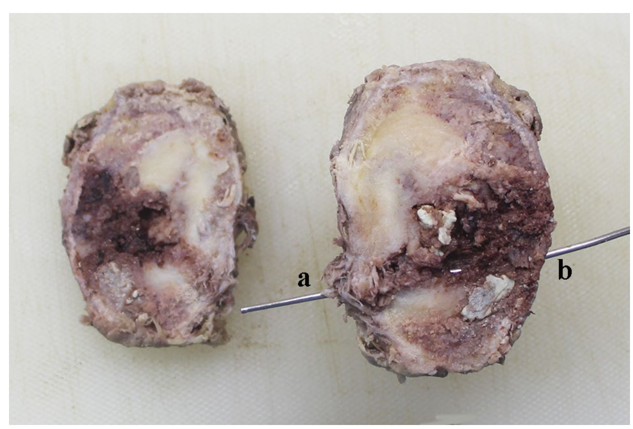

Fig. 4 **a** L5 pedicle (wound entry). **b** L5 vertebral body (wound exit). The probe was passed through the L5 pedicle and the L5 vertebral body. The intervertebral disc between L4 and L5 was observed to be relatively complete, and a surgical wound was seen on the transverse section. The wound passed through part of the intervertebral disc from the pedicle of L5 and finally penetrated the L5 vertebral body. Around the wound, there was an irregular, milky yellow bone cement

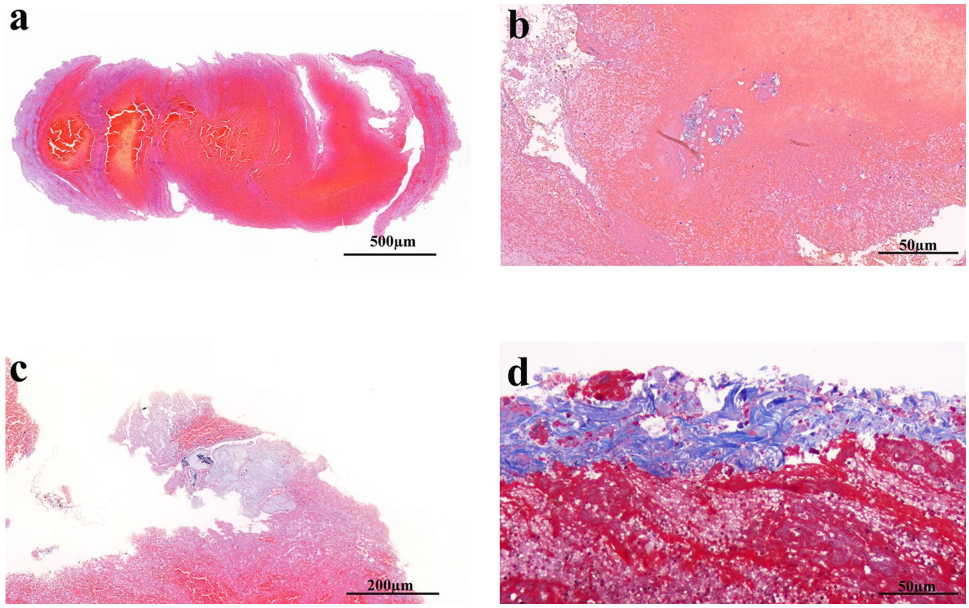

Fig. 5 Histological pictures (hematoxylin-eosin staining and Masson staining) of the lesions. **a** The pulmonary artery and bilateral pulmonary hilar thromboses were mixed thrombi. **b** Chondrocytes in the pulmonary hilar thromboembolism. **c** Calcium deposits (bone) and collagen fibers (fascia) in the pulmonary hilar thromboembolism. **d** Blue collagen fibers were seen in the hilar thromboembolism (Masson staining)

enters the venous system, 5–16.6% of which enters the prevertebral vein, and 16.5% of which enters the epidural vein. However, there are a few reports of bone cement leaking into the inferior vena cava [8]. The underlying mechanism of bone cement leakage is associated with insufficient polymerization of the injected bone cement, overfilling of the vertebral

Table 1 Eight case reports of deaths from complications after percutaneous vertebroplasty

Author/Publication Year	Age (years old)/sex	Primary Disease	Medical history	Postoperative death time	Autopsy	Cause of death
Chen/2002 [9]	75/Female	Osteoporosis with compression fractures of T12, L2 and L4	Hypertension, minor diastolic dysfunction and pulmonary hypertension	Unknown	N	Pulmonary fat embolization
Yoo/2004 [10]	68/Female	Osteoporotic compression fracture of L5	No significant medical or family history	20d	Unknown	Pulmonary cement embolism
Stricker/2004 [11]	83/Female	An osteoporotic fracture of first lumbar vertebra	Chronic obstructive pulmonary disease, recurrent congestive heart failure	11d	Y	Pulmonary cement embolism
Monticelli/2005 [12]	81/Female	A fracture of the first lumbar vertebra	Unknown	15min	Y	Pulmonary cement embolism
Syed/2006 [13]	69/Female	Acute/subacute compression fractures of T7–T12	Emphysema (requiring home oxygen), pulmonary hypertension with cor pulmonale, acid reflux, coronary artery disease, cardiac arrythymias, and hypertension	2.58h	Y	Pulmonary fat embolization
Marden/2008 [14]	71/Female	T12 compression fracture	Unknown	24h	N	Brain cement embolism
Lee, MRCP/2014 [15]	Unknown	Osteoporotic fracture	Unknown	3months	Unknown	Cardiac tamponade, acute respiratory distress syndrome
D'Errico /2019 [16]	72/Female	Multiple thoracic vertebral compression fractures	Reduction mastoplasty	3 h	Y	Cardiac tamponade

body, and vein damage during PVP; when cement leaks into veins, it has the capacity to activate the coagulation system.

Pulmonary thromboembolism associated with PVP is a rare complication in clinical and forensic practice. To date, few cases of fatal pulmonary thromboembolism post-PVP have been reported. We searched "complications or death after percutaneous vertebroplasty" in PubMed and Web of Science, excluding only abstracts and non-English articles, and we carefully analyzed 8 identified studies (Table 1) [9–16]. All the affected victims were older than 65 years, and most of them had undergone PVP due to vertebral compression fractures. Bone cement embolism and fat embolism were the leading types of embolisms associated with PVP. Death related to PVP can occur within 1 day to 3 months later. In addition, due to the PVP operation, the limited activities post-surgery and the long period of bedrest may contribute to deep venous thrombosis formation in the lower extremities and thus lead to pulmonary thromboembolism.

Here, the patient was older, the surgical duration was long, and she was bedridden after the operation. Even without other risk factors for deep venous thrombosis, the possibility of deep venous thrombosis in such cases is still high. Since two operations were performed on L5, it is necessary to determine the source of injury. According to the difference in surgical methods and surgical sites and the autopsy results, we theorize that injury to the inferior vena cava was caused by PVP. The second operation may have played a facilitating role in the PTE.

The present patient displayed PVP-induced inferior vena cava injury resulting in fatal pulmonary thromboembolism formation. It is well known that such emboli are usually composed of off-white bone cement. Here, brown or black granules were found in the pulmonary arteries or small blood vessels under a microscope. After a detailed autopsy and histopathological examination, bone cement embolism post-PVP was excluded. In addition, the rather uncommon components in the pulmonary thromboembolism and deep venous thrombosis confirmed the common origin.

In forensic practice, Prussian blue staining, Masson staining, and other methods can be used to analyze the material composition of a thrombus to estimate the injury time. Unfortunately, Prussian blue staining was negative in the present case. Blue collagen fibers were seen only in the hilar thromboembolus but not in the inferior vena cava thrombosis. Therefore, this staining method only played a supporting role in the conclusions made for this case.

Conclusion

PVP is a rare cause of pulmonary thromboembolism that should attract the attention of clinical physicians and forensic pathologists. In such cases, forensic pathologists should perform a detailed autopsy and find the link between the surgery and the fatal outcome.

Key points

1. Fatal pulmonary thromboembolism following PVP is rare in medical practice.
2. The cause of pulmonary thromboembolism may be inferior vena cava injury during percutaneous vertebroplasty.
3. A detailed autopsy and careful histological examination should be carried out to analyze the origin of a pulmonary thromboembolism.
4. Forensic pathologists should find the link between the medical history and the fatal outcome.

Declarations

Informed consent Written informed consent for publishing this scientific report was obtained from the direct relative of the decedent in this case.

Conflict of interest The authors declare that they have no conflict of interest.

References

1. Hoyt D, Urits I, Orhurhu V, Orhurhu MS, Callan J, Powell J, Manchikanti L, Kaye AD, Kaye RJ, Viswanath O. Current concepts in the management of vertebral compression fractures. Curr Pain Headache R. 2020;24(5). https://doi.org/10.1007/s11916-020-00849-9.
2. Buchbinder R, Johnston RV, Rischin K, Homik J, Jones CA, Golmohammadi K, Kallmes DF. Percutaneous vertebroplasty for osteoporotic vertebral compression fracture (review). Wiley. 2018;4(4):CD006349. https://doi.org/10.1002/14651858.CD006349.pub3.
3. Hsieh M, Kao F, Chiu P, Chen L, Yu C, Niu C, Lai P, Tsai T. Risk factors of neurological deficit and pulmonary cement embolism after percutaneous vertebroplasty. J Orthop Surg Res. 2019;14(406). https://doi.org/10.1186/s13018-019-1459-4.
4. Allen W, Burton. Basem Hamid. Kyphoplasty and vertebroplasty. Curr Pain Headache Rep. 2008;12(1):22–7. https://doi.org/10.1007/s11916-008-0005-6.
5. Fehlings MG. The safety of percutaneous vertebroplasty and kyphoplasty. J Neurosurg Spine. 2009;11(5):605–6. https://doi.org/10.3171/2009.6.SPINE09359.
6. Hafezi-Nejad N, Bailey CR, Solomon AJ, Abou Areda M, Carrino JA, Khan M, Weiss CR. Vertebroplasty and kyphoplasty in the USA from 2004 to 2017: national inpatient trends, regional variations, associated diagnoses, and outcomes. J Neurointerv Surg. 2021;13(5):483–91. https://doi.org/10.1136/neurintsurg-2020-016733.
7. Wang L, Yang H, Shi Y, Jiang W, Chen L. Pulmonary cement embolism associated with percutaneous vertebroplasty or kyphoplasty: a systematic review. Orthop Surg. 2012;4(3):182–9. https://doi.org/10.1111/j.1757-7861.2012.00193.x.
8. Laredo J, Hamze B. Complications of percutaneous vertebroplasty and their prevention. Seminars in Ultrasound, CT and MRI. 2005;26(2):65–80. https://doi.org/10.1053/j.sult.2005.02.003.
9. Chen HL, Wong CS, Ho ST, Chang FL, Hsu CH, Wu CT. A lethal pulmonary embolism during percutaneous vertebroplasty. Anesth Analg. 2002;95(4):1060–2. https://doi.org/10.1213/01.ANE.0000025591.86476.F2.

10. Yoo KY, Jeong SW, Yoon W, Lee J. Acute respiratory distress syndrome associated with pulmonary cement embolism following percutaneous vertebroplasty with polymethylmethacrylate. Spine (Phila Pa 1976). 2004;29(14):E294–7. https://doi.org/10.1097/01.brs.0000131211.87594.b0.
11. Stricker K, Orler R, Yen K, Takala J, Luginbühl M. Severe hypercapnia due to pulmonary embolism of polymethylmethacrylate during vertebroplasty. Anesth Analgesia. 2004;98(4):1184–6. https://doi.org/10.1213/01.ANE.0000104585.83801.C5.
12. Monticelli F, Meyer HJ, Tutsch-Bauer E. Fatal pulmonary cement embolism following percutaneous vertebroplasty (PVP). Forensic Sci Int. 2005;149(1):35–8. https://doi.org/10.1016/j.forsciint.2004.06.010.
13. Syed MI, Jan S, Patel NA, Shaikh A, Marsh RA, Stewart RV. Fatal fat embolism after vertebroplasty: identification of the high-risk patient. AJNR Am J Neuroradiol. 2006;27(2):343–5.
14. Marden FA, Putman CM. Cement-embolic stroke associated with vertebroplasty. Am J Neuroradiol. 2008;29(10):1986–8. https://doi.org/10.3174/ajnr.A1159.
15. Lee V, PatelL R, Meier P, Lawrence D, Roberts N. Conservative management of inferior vena cava cement spike after percutaneous vertebroplasty causes fatal cardiac tamponade. J Rhuematol. 2014;41(1):141–2. https://doi.org/10.3899/jrheum.130570.
16. D'Errico S, Niballi S, Bonuccelli D. Fatal cardiac perforation and pulmonary embolism of leaked cement after percutaneous vertebroplasty. J Forensic Leg Med. 2019;63:48–51. https://doi.org/10.1016/j.jflm.2019.03.004.